

GEN-3 SAFE HELICOPTER

DOCUMENT 00

10-TON / 7-ROTOR SAFETY-CENTERED BASELINE

Central Protected Fixed Safety Capsule Configuration

Configuration: Seven equally sized 6.0 m rotors distributed around the central capsule; no primary lift rotor directly above the occupied capsule volume.

Landing architecture: Seven external mechanical damper landing legs with energy-absorbing elements; six additional capsule-side isolation damper elements are separate from the seven landing legs.

Control Item	Value
Project	GEN-3 SAFE HELICOPTER
Platform	GEN-3 Safe 10-Ton Modular VTOL Helicopter
Founder / Principal Author	Amirbahador Atai
Configuration ID	GEN3-SH-10T-FC
Baseline Revision	Rev A
Document Revision	00-T10-R7-9.7
Classification	Confidential
Program Status	Engineering Design / Concept-Pre-Feasibility
Certification Status	Not certified; compliance not established

Engineering maturity: CONCEPT / PRELIMINARY PRE-FEASIBILITY. This document is a controlled design baseline, not a substantiation package, type certificate, flight manual, production drawing set, or verified performance record.

Document-quality target: 9.7/10 for configuration discipline, evidence control, traceability, and engineering-document quality. This score is an editorial/process assessment, not proof of aircraft feasibility, safety, certification, or performance.

2. Document Control and Revision Record

Controlled engineering record. Evidence status governs claim strength.

Item	Controlled Value	Status
Document Number	00	[V] document control
Document Revision	00-T10-R7-9.7	[T] controlled document revision
Aircraft Configuration Revision	Rev A	Controlled baseline
Configuration ID	GEN3-SH-10T-FC	Controlled
Parent Baseline	Document 00 / prior optimized baseline	Controlled source
Purpose	Master platform baseline and engineering-control reference	[AR]
Classification	Confidential	Controlled
Engineering maturity	Concept / pre-feasibility	Evidence-supported
Change principle	No silent baseline change	Mandatory control rule

Revision intent. This revision corrects the landing-gear terminology and locks the external landing architecture at seven mechanical damper legs. The six capsule-side damper elements remain part of the capsule isolation chain and are not counted as external landing legs.

Important distinction. Document revision and aircraft configuration revision remain separate. Editing this document does not by itself change the aircraft configuration.

Controlled change requirement. Any future baseline change requires a Change ID, current value, proposed value, source, rationale, affected requirements, mass/performance/safety/interface/verification impact, affected documents, authority and approval status.

3. Executive Summary

Controlled engineering record. Evidence status governs claim strength.

The Gen-3 baseline is a compact 10-ton VTOL concept organized around a centrally protected fixed safety capsule. The seven primary rotors are equal in nominal diameter and distributed around the capsule. The architecture intentionally removes a primary lift rotor from directly above the occupied capsule volume.

The safety concept is layered rather than dependent on one emergency mechanism. The baseline combines a protected capsule, multi-stage mechanical energy absorption, distributed lift channels, multiple prime movers, fire containment/isolation, flight-control fault management, emergency recovery concepts, and water-ditching provisions.

Baseline item	Controlled value	Status
Design mass	10,000 kg target	[T]
Rotor count	7	[T] / controlled architecture
Rotor diameter	6.0 m each	[T]
Rotor arrangement	Distributed around central capsule; no roof rotor over capsule	[T]
Envelope	17 m × 17 m × 5 m	[T]
Main body height	4 m nominal	[T]
Capsule rise	≈1 m above main body	[T]
Fuel capacity	1,700 L nominal	[T]
Prime movers	4 turboshaft-class	[T]
Installed power	2.5–2.8 MW target	[T]
External landing legs	7 mechanical damper legs	[T]
Capsule-side isolation dampers	6	[T]
Occupants	7	[T]

Claim boundary: No survivability, crashworthiness, fire-survivability, water-survivability, fail-safe, airworthiness, certification, or production-readiness claim is established by this document.

4. Controlled Baseline

Controlled engineering record. Evidence status governs claim strength.

Parameter	Baseline	Status
Primary rotors	7	[T] Controlled
Nominal rotor diameter	6.0 m	[T] Controlled
Total geometric disk area	$\approx 197.9 \text{ m}^2$	[C] from $7 \times \pi(3 \text{ m})^2$
Disk loading at 10,000 kg	$\approx 50.5 \text{ kg/m}^2$	[C] planning calculation
Overall length	17 m	[T]
Overall width	17 m	[T]
Overall height	5 m	[T]
Main body height	4 m	[T]
Capsule rise above body	$\approx 1 \text{ m}$	[T]
Fuel tank capacity	1,700 L	[T]
Reference fuel mass	$\approx 1,360 \text{ kg}$ at 0.80 kg/L	[C-A] density assumption
Prime movers	4 × turboshaft-class	[T]
Power class per unit	$\approx 650\text{--}700 \text{ kW}$ class	[T]
Installed power target	2.5–2.8 MW	[T]
Occupant capacity	7	[T]
Capsule	Fixed / non-jettisonable	[T]
External landing legs	7 mechanical damper legs	[T]
Capsule-side damper elements	6 mechanical isolation elements	[T]
Recovery concept	8-parachute aircraft-level concept	[T]

5. Rotor Architecture and Geometry Control

Controlled engineering record. Evidence status governs claim strength.

The rotor architecture is frozen at seven equally sized nominal 6.0 m rotors. The central occupied capsule is surrounded by distributed rotor stations. There is no primary lift rotor directly above the capsule roof line in the controlled baseline.

Parameter	Value	Control
Number of rotors	7	Locked baseline
Diameter each	6.0 m nominal	Locked target
Area each	≈28.27 m ²	Calculated
Total geometric disk area	≈197.9 m ²	Calculated
Central roof rotor	None	Locked architectural feature
Exact azimuthal station coordinates	[AV]	Requires controlled geometry definition
Blade count	[AV]	Requires rotor selection
Solidity	[AV]	Requires aerodynamic model
Tip speed	[AV]	Requires rotor design
Collective/cyclic architecture	[AV]	Requires control concept

Consistency flag: The phrase “distributed around the capsule” defines architecture but does not yet constitute a fully dimensioned rotor-location drawing. Exact rotor center coordinates and clearance envelopes remain [AV].

Engineering consequence: Seven equal rotors simplify configuration control and symmetry of hardware families, but distributed placement must be validated for aerodynamic interference, control authority, structural load transfer, noise, maintenance access and failure reconfiguration.

6. Central Fixed Safety Capsule

Controlled engineering record. Evidence status governs claim strength.

Feature	Baseline concept	Status
Location	Central heart of vehicle	[T]
Vertical position	≈1 m above main body	[T]
Separation from roof rotor	No primary lift rotor directly above capsule	[T]
Primary shell concept	Rounded / multi-directional impact load management	[T]
Internal reinforcement	Multi-angle carbon-fiber roll-cage concept	[T]
Seat concept	Soft damper-mounted seats	[T]
Capsule-to-airframe interface	Multiple high-stroke mechanical isolation mounts	[T]
Capsule-side damper elements	6	[T]
Jettison capability	None	[T]
Independent flight capability	None	[T]

The capsule is treated as a system-level occupant-protection architecture rather than simply a stronger cabin. The design intent is to manage load paths, deformation, energy absorption, restraint loads, fire separation and egress as a coupled system.

Verification boundary: No crash-survivability performance level is claimed. Required evidence includes structural analysis, material characterization, full-scale or representative dynamic tests, seat/restraint tests, egress assessment, and hazard-specific verification.

7. Seven Mechanical Landing Legs

Controlled engineering record. Evidence status governs claim strength.

The external landing architecture is explicitly controlled at seven mechanical damper legs. This is a correction from prior wording that could be read as a six-leg baseline. The seven external legs are distinct from the six capsule-side isolation damper elements.

Parameter	Controlled baseline	Status
External landing legs	7	[T]
Mechanical actuation / energy absorption	Passive/mechanical damper concept	[T]
Leg tip concept	Thick energy-absorbing contact element	[T]
Primary function	Ground load path + energy absorption	[T]
Leg attachment geometry	[AV]	Detailed structural layout required
Stroke	[AV]	Dynamic load analysis required
Force-displacement curve	[AV]	Damper characterization required
Ultimate load	[AV]	Structural substantiation required
Ground contact loads	[AV]	Landing cases required
Redundancy / fail-safe behavior	[AV]	Safety analysis required

Design rule: Do not assign survival credit to a landing leg merely because it contains a damper. Survival credit requires defined load cases, energy capacity, stroke, attachment strength, failure behavior, and validated occupant-load transfer.

8. Capsule Isolation and Energy-Absorption Chain

Controlled engineering record. Evidence status governs claim strength.

The intended energy-management chain is: external ground interface → seven landing-leg responses → primary airframe load paths → capsule support stroke → six capsule-side damper elements → capsule shell/roll-cage → seat/restraint attenuation.

Element	Purpose	Evidence required
7 external legs	Ground-impact energy absorption	Dynamic test + structural analysis
Primary airframe	Controlled load distribution/deformation	FEA + physical load testing
Capsule mounts	Isolation / stroke management	Characterization + dynamic test
6 capsule-side dampers	Secondary isolation stage	Component and integrated testing
Roll-cage	Occupant-space load-path control	FEA + material/coupon + structural test
Seats/restraints	Occupant energy attenuation	Seat/restraint test program

Open issue: Stroke, peak force, damping law, temperature dependence, rate dependence and failure mode of each energy-absorbing element are [AV]. These values must be derived from defined impact scenarios rather than selected to fill a table.

9. Propulsion Architecture

Controlled engineering record. Evidence status governs claim strength.

Item	Baseline	Status
Prime movers	4 turboshaft-class	[T]
Power class	≈650–700 kW class per unit	[T]
Installed aircraft power target	2.5–2.8 MW	[T]
Energy conversion concept	Turboshaft → generator → segmented HV bus → inverter → rotor motor	[T]
Rotor channels	7 independent drive channels	[T]
Isolation principle	Faulty channel can be isolated	[T]
Exact engine OEM/model	[AV]	Direct OEM engagement required
Generator sizing	[AV]	Matching study required
Thermal rejection	[AV]	Thermal model required
Electrical architecture	[AV]	Detailed system design required

The architecture is intentionally described as a direction rather than a completed propulsion design. Shared vulnerable buses or common cooling systems cannot be counted as independent safety channels without common-cause analysis.

Major verification item: Single-prime-mover and single-rotor-channel loss cases require validated control reconfiguration and sufficient remaining power/authority under the defined mission and environment.

10. Preliminary Power and Hover Sanity Checks

Controlled engineering record. Evidence status governs claim strength.

The following are engineering sanity checks, not demonstrated aircraft performance. They use only the controlled mass, rotor geometry and explicitly stated assumptions.

Quantity	Preliminary result	Status / limitation
Weight at 10,000 kg	≈98.1 kN	[C] using $g = 9.81 \text{ m/s}^2$
Thrust per rotor at equal sharing	≈14.0 kN	[C-A] assumes equal total-thrust sharing
Illustrative 20% total thrust margin	≈117.7 kN total	[C-A] design illustration only
Illustrative per-rotor thrust with margin	≈16.8 kN	[C-A]
Ideal induced hover power floor	≈1.40 MW	[C-A] idealized momentum-theory basis
Approx. real hover power	≈1.94 MW	[C-A] uses $FM \approx 0.72$; requires validation
Installed power target	2.5–2.8 MW	[T] not a demonstrated margin

Critical distinction: The 1.94 MW value is not a certified hover requirement, and the 2.5–2.8 MW target is not proof of hover margin. Profile power, transmission/electrical losses, hot/high conditions, control margins, climb, reserve power, thermal limits and degraded operation remain open.

11. Fuel and Energy Management

Controlled engineering record. Evidence status governs claim strength.

Parameter	Current baseline	Status
Fuel capacity	1,700 L	[T]
Reference density	0.80 kg/L	[A] planning assumption
Reference fuel mass	≈1,360 kg	[C-A]
Fuel type	[AV]	Final propulsion/OEM definition required
Tank segmentation	[AV]	Hazard and CG analysis required
Fire isolation	Required concept	[AR/T]
Fuel shutoff	Required concept	[AR/T]
Tank structural qualification	[AV]	Test program required
Endurance	[AV] until mission profile is frozen	Do not use prior estimate as guarantee
Range	[AV] until aerodynamic/mission model is closed	Do not use prior estimate as guarantee

The previous preliminary endurance/range figures are deliberately not frozen into the controlled baseline. A defensible range/endurance claim requires mission definition, aircraft mass closure, propulsion SFC, aerodynamic drag, reserve policy, environmental conditions and verified installed-system performance.

12. Preliminary Mass Control

Controlled engineering record. Evidence status governs claim strength.

Mass group	Planning status	Closure
Primary airframe / rotor support	≈2,050 kg planning allocation	[A]
Safety capsule / occupant protection	≈1,150 kg planning allocation	[A]
7 rotors / hubs / control hardware	≈720 kg planning allocation	[A]
Powerplant / generators / motors / drives	≈1,550 kg planning allocation	[A]
Fuel system excluding fuel	≈230 kg planning allocation	[A]
Fuel at 1,700 L	≈1,360 kg planning value	[C-A]
Avionics / flight control / distribution	≈520 kg planning allocation	[A]
Landing gear: 7 external + 6 capsule elements	≈450 kg planning allocation	[A]
Fire / emergency / safety equipment	≈540 kg planning allocation	[A]
Thermal management	≈350 kg planning allocation	[A]
Interconnects / hardware / fluids / growth	≈1,180 kg planning allocation	[A]
Total planning mass	≈9,800 kg	[C-A] planning only
Target	10,000 kg	[T]

Mass status: PARTIAL / OPEN. The planning total does not constitute a mass closure. Component-level supplier data, detailed structural sizing, rotor/hub selection, thermal hardware, electrical conversion hardware, emergency equipment and weighed test articles are required.

13. Flight-Control and Fault-Reconfiguration Concept

Controlled engineering record. Evidence status governs claim strength.

The seven-rotor architecture requires a control system able to maintain controllability under rotor-channel and prime-mover faults. This is a design objective, not a verified capability.

Function	Concept	Verification needed
Rotor health monitoring	Speed, power, temperature, vibration, actuator state	HIL + bench + flight test
Fault detection	Independent monitoring paths where practical	Fault-injection test
Fault isolation	Automatic channel isolation	Integrated test
Control reallocation	Recompute rotor commands within validated limits	HIL + flight
Emergency power	Independent power for flight-critical functions	Electrical analysis + test
Common-mode control	Segregation of sensing/computation/actuation	FMEA/FMECA/SSA
Handling qualities	Defined by mission and certification basis	Flight test

Open technical issue: Exact rotor station geometry, rotor control authority, actuator limits, transient response and failure combinations must be defined before claiming single-failure controllability.

14. Structural Load-Path Philosophy

Controlled engineering record. Evidence status governs claim strength.

The primary structure is organized as a perimeter around the capsule, with multiple load paths intended to separate rotor, propulsion, landing and emergency loads where practical.

Load case family	Current status	Required work
Rotor thrust / torque	[AV]	Define rotor loads and interfaces
Prime-mover loads	[AV]	Engine/generator installation data
Landing loads	[AV]	7-leg load cases + ground conditions
Capsule isolation loads	[AV]	Damper force/stroke definition
Parachute attachment loads	[AV]	Recovery system sizing
Water-impact loads	[AV]	Hydrodynamic/structural modelling
Fire / thermal gradients	[AV]	Thermal-structural analysis
Ultimate factors	[AV]	Applicable certification basis pending

No material allowables, laminate schedules, structural thicknesses or ultimate load factors are frozen here. Those values require controlled material sources, structural sizing and the eventual certification basis.

15. Fire, Thermal and Hazard Containment

Controlled engineering record. Evidence status governs claim strength.

Fire safety is treated as a containment-and-isolation problem. The protected capsule is separated from high-energy machinery by dedicated zones and barriers in the conceptual architecture.

- Independent detection logic where practical.
- Fuel isolation and electrical isolation following hazard analysis.
- Fire-resistant barriers between hot machinery and occupied space.
- Suppression around capsule perimeter and critical high-energy zones.
- Thermal monitoring of engines, generators, fuel system and power electronics.
- Emergency power reduction and load shedding.
- Post-fire safe-state logic.

Evidence boundary: No fire-survivability claim is established. Suppression agent, quantity, discharge time, compartment leakage, thermal limits, smoke/toxicology, and post-fire structural effects remain [AV].

16. Emergency Recovery Concept

Controlled engineering record. Evidence status governs claim strength.

Scenario	Primary concept	Secondary concept	Status
Single rotor failure	Fault reconfiguration	Landing mode	[T]
Single prime-mover failure	Power/control redistribution	Landing mode	[T]
Multiple-channel degradation	Degraded flight mode	Aircraft recovery system	[T]
Major fire	Isolation + power reduction	Emergency recovery / landing	[T]
Severe structural event	Maintain capsule integrity	Recovery concept	[T]
Water impact	Capsule protection + egress	Flotation / recovery	[T]

The baseline contains an aircraft-level concept for eight parachutes. The concept is not sized or verified. Deployment loads, minimum deployment altitude, failure modes, canopy interactions, attachment structure, descent rate and landing energy remain [AV].

Safety rule: Emergency recovery devices receive no credited survival benefit until their complete load path and deployment sequence are analytically and physically substantiated.

17. Water Survival and Egress

Controlled engineering record. Evidence status governs claim strength.

The fixed capsule is intended to preserve a protected occupant volume during ditching. The concept includes flotation, protected egress and retention of emergency equipment, but no water-impact survivability is yet demonstrated.

Topic	Status	Required evidence
Flotation stability	[AV]	Hydrostatic / dynamic analysis + test
Capsule leakage	[AV]	Representative water testing
Emergency exits	[T] design objective	Human-factors + practical egress test
Inverted orientation	[AV]	Ditching/egress test
Emergency equipment retention	[AV]	Impact and water tests
Electrical isolation	[AV]	System safety test

No numerical flotation capacity, ditching angle, water-entry speed or survival duration is fabricated in this baseline.

18. Human Factors and Occupant Protection

Controlled engineering record. Evidence status governs claim strength.

Occupant protection is a system-level problem involving restraint, seat response, capsule deformation, egress, fire/smoke, visibility, controls, emergency equipment and human tolerance.

Area	Current status	Closure
Occupant count	7	[T]
Seat concept	Soft damper-mounted	[T]
Restraint system	[AV]	Occupant-protection design
Head/neck protection	[AV]	Biomechanical analysis/test
Emergency egress	Multiple routes intended	[T] / [AV]
Smoke protection	[AV]	Fire/human-factors analysis
Cabin access	[AV]	Human-factors layout
Emergency lighting	[AV]	Systems definition

Boundary: No occupant injury criterion or crash-survivability number is asserted without a defined regulatory/engineering basis and test method.

19. Verification, Validation and Test Strategy

Controlled engineering record. Evidence status governs claim strength.

Subsystem	Analysis	Bench / Rig	Dynamic / Flight	Status
Rotor / hub	Yes	Yes	Yes	PLANNED
Propulsion channel	Yes	Yes	Yes	PLANNED
Flight controls	Yes	HIL + fault injection	Yes	PLANNED
Capsule structure	FEA	Full-scale article	Impact / flight correlation	PLANNED
7 external landing legs	Load model	Component characterization	Full-scale dynamic	PLANNED
6 capsule isolation dampers	Dynamic model	Component characterization	Integrated	PLANNED
Fire protection	Hazard analysis	System fire test	As required	PLANNED
Emergency recovery	Simulation	Deployment rig	Flight demonstration	PLANNED
Water survival	Hydro / structural model	Water test	As required	PLANNED

Verification method does not equal verification result. All entries above are planned unless supported by documented evidence.

20. Requirements and Traceability Framework

Controlled engineering record. Evidence status governs claim strength.

Document 00 establishes architecture and control rules but does not replace a formal requirements database. Formal SHALL statements should be allocated only after authority and source are identified.

Requirement ID	Requirement statement	Source	Verification	Status
REQ-ARCH-001	The aircraft SHALL use seven primary lift rotors in the controlled baseline	Baseline	Inspection / configuration audit	[AR]
REQ-ARCH-002	The seven baseline rotors SHALL have equal nominal diameter of 8.0m	Program baseline	Inspection	[AR]
REQ-ARCH-003	The controlled baseline SHALL contain no primary lift rotor directly above the occupied capsule volume	Program baseline	Inspection	[AR]
REQ-GND-001	The external landing architecture SHALL contain seven mechanical dampers	Program baseline	Inspection + test	[AR]
REQ-CAPS-001	The occupant capsule SHALL remain mechanically integrated and Program baseline in Variant A.	Program baseline	Inspection	[AR]
REQ-SAFE-001	Safety claims SHALL be supported by defined evidence before verification control rule	Engineering control	Audit	[AR]

These entries are traceability scaffolding derived from the controlled baseline. The eventual requirements management system should assign authoritative IDs, owners, rationale and approval status.

21. Configuration Management and Change Control

Controlled engineering record. Evidence status governs claim strength.

The engineering package is treated as one configuration-controlled system. All dependent documents shall reference the same baseline identifiers, parameter IDs, requirement IDs and evidence IDs.

Change field	Required content
Change ID	Unique controlled identifier
Current value	Existing baseline value
Proposed value	Proposed change; not baseline until approved
Reason	Engineering or program rationale
Source	Controlled source / external source / calculation / test
Requirements impact	Affected IDs
Mass impact	Updated mass statement
Performance impact	Updated models/analysis
Safety impact	Hazard and safety assessment
Interface impact	Affected physical/data/power interfaces
Verification impact	New or repeated evidence
Affected documents	Downstream document list
Authority	Approver / approval board
Approval status	Pending / approved / rejected

Rule: Proposed values are not part of the controlled baseline until formally approved.

22. Source and Data Provenance Control

Controlled engineering record. Evidence status governs claim strength.

Class	Meaning	Permitted use
[V]	Verified engineering evidence	May support factual claim
[V-TEST]	Verified by documented test	Test-backed claim
[V-INSP]	Verified by inspection	Configuration/physical confirmation
[V-AN]	Verified by accepted analysis	Analysis-backed claim
[C]	Calculated from supported inputs	Derived value
[C-A]	Calculated using assumptions	Planning only; assumption visible
[C-SIM]	Simulation/model result	Model result, not test
[A]	Engineering assumption	Temporary planning input
[T]	Design target	Intent, not achievement
[AR]	Allocated requirement	Requirement control
[AV]	Awaiting definition/validation/verification	Open
[ND]	Not determined	Open
[OPEN]	Unresolved/conflicting	Do not reconcile silently

External information may be used only when explicitly identified as external. Unverified citations, standards, regulations or supplier claims shall not be presented as verified evidence.

23. Engineering Consistency Audit

Controlled engineering record. Evidence status governs claim strength.

Check	Finding	Status	Required action
Rotor count	7 consistently stated	PASS	Maintain
Rotor diameter	6.0 m all seven	PASS	Maintain
Central rotor	No primary roof rotor	PASS	Maintain
Envelope	17 × 17 × 5 m; body 4 m; capsule +1 m	PASS / geometry pending	Freeze CAD geometry
External legs	Corrected to 7 mechanical damper legs	PASS	Maintain
Capsule dampers	6 separate isolation elements	PASS	Keep separate from leg count
Fuel capacity	1,700 L	PASS	Maintain
Prime movers	4 turboshaft-class	PASS	OEM matching pending
Installed power	2.5–2.8 MW target	PASS as target	Do not treat as verified
Mass	10 t target; planning mass only	OPEN	Component mass closure
Hover power	1.94 MW [C-A]	OPEN	Detailed rotor/system model
Endurance/range	Not frozen	OPEN	Mission + aero + propulsion model
Safety architecture	Conceptual	OPEN	FHA/FMEA/FTA/PSSA/SSA

Engineering consistency flag [OPEN]: the baseline is internally coherent at architecture level, but several performance and safety assertions remain dependent on missing engineering evidence. No contradiction is silently resolved.

24. Risk Register and Closure Logic

Controlled engineering record. Evidence status governs claim strength.

Risk / issue	Potential impact	Current treatment	Closure evidence
Mass growth above 10 t	High	Planning reserve	Supplier data + weighed hardware
Insufficient installed power	High	2.5–2.8 MW target	Rotor/propulsion/thermal model
7-leg stroke insufficient	High	Mechanical damper concept	Dynamic impact test
Control authority without roof rotor	Medium-High	Distributed layout	6-DOF model + HIL + flight
Common-mode control failure	High	Segregated architecture objective	FMEA/FMECA + SSA + fault injection
Fire propagation	High	Containment/isolation concept	Hazard analysis + fire tests
Recovery-system load exceedance	High	8-parachute concept	Deployment simulation + structural test
Water-impact instability	High	Capsule protection concept	Hydrodynamic model + water test
Engine/generator mismatch	High	OEM engagement	Matching study + supplier data

If no approved project risk matrix exists, numerical risk scores are not assigned. The table identifies engineering closure needs rather than pretending that risk has been reduced to an accepted level.

25. Manufacturing, Prototype and Development Strategy

Controlled engineering record. Evidence status governs claim strength.

The compact 10-ton baseline is intended to support staged physical development. The following is a development sequence, not a completed manufacturing capability.

Phase	Objective	Evidence gate
1	Material/coupon and joint tests	Material and process data
2	Subscale rotor/motor/controller/thermal rigs	Component performance
3	Full-scale capsule + 7-leg impact articles	Energy absorption evidence
4	Integrated propulsion test rig	Power/thermal/control evidence
5	Unmanned integrated demonstrator	Integrated system evidence
6	Progressive flight-envelope expansion	Flight-test evidence
7	Certification-oriented conformity/compliance program	Regulatory evidence

Manufacturing process capability, tooling, production rate, unit cost and supply-chain maturity remain [AV] unless separately supported by controlled evidence.

26. Engineering Maturity and Claim Boundary

Controlled engineering record. Evidence status governs claim strength.

Maturity level	Current applicability
CONCEPT	Yes — current evidence level
PRELIMINARY DESIGN	Not yet fully established
DETAILED DESIGN	No
VERIFICATION	No

Evidence-supported maturity: CONCEPT / PRE-FEASIBILITY.

The document is intentionally more disciplined than the underlying evidence. That is desirable for configuration control, but documentation sophistication must not be mistaken for engineering maturity.

Claim boundary. The following remain unproven unless separate evidence is accepted: crash survivability, fire survivability, water survivability, guaranteed degraded-flight capability, fail-safe status, airworthiness, certification, production readiness, patentability, novelty, freedom to operate and legal protection.

Permitted wording: “design concept,” “engineering target,” “preliminary calculation,” “planned verification,” “candidate safety architecture,” and “controlled baseline.”

27. Quality Gate and Release Checklist

Controlled engineering record. Evidence status governs claim strength.

Gate	Release criterion	Result
Baseline	7 rotors; all 6.0 m	PASS
Baseline	17 × 17 × 5 m; body 4 m; capsule +1 m	PASS
Baseline	10,000 kg target	PASS
Baseline	1,700 L target fuel capacity	PASS
Baseline	4 turboshaft-class prime movers; 2.5–2.8 MW target	PASS
Baseline	7 external mechanical damper legs	PASS
Capsule	Fixed / non-jettisonable; centrally protected	PASS
Rotor protection	No primary lift rotor directly above capsule	PASS
Integrity	No fabricated verification/test results	PASS
Integrity	Targets separated from calculated values	PASS
Traceability	Requirements/evidence architecture identified	PASS / PARTIAL
Configuration	Document revision separated from configuration revision	PASS
Uncertainty	Missing values marked [AV]/[ND]/[OPEN]	PASS
Maturity	Concept/pre-feasibility stated	PASS
Safety	Safety claims explicitly bounded	PASS
Performance	No prior endurance/range frozen as fact	PASS

Release disposition: ACCEPTED AS A CONTROLLED CONCEPT BASELINE, subject to the open engineering items listed in this document. Not accepted as evidence of feasibility, certification, airworthiness or demonstrated safety.

28. Final Baseline Statement and Engineering Status

Controlled engineering record. Evidence status governs claim strength.

Document 00 Rev A establishes the controlled Gen-3 Safe Helicopter 10-ton fixed-capsule baseline: seven equally sized nominal 6.0 m rotors distributed around a centrally protected capsule; no primary lift rotor directly above the occupied capsule; overall target envelope 17 m × 17 m × 5 m with a nominal 4 m main body and approximately 1 m capsule rise; 1,700 L nominal fuel capacity; four turboshaft-class prime movers with a 2.5–2.8 MW installed-power target; seven external mechanical damper landing legs; six separate capsule-side isolation damper elements; seven occupants; and a ten-layer safety architecture.

The corrected landing-gear baseline is explicitly seven external mechanical damper legs. The six capsule-side dampers are not counted as landing legs and remain a separate capsule-isolation subsystem.

The central engineering proposition is layered occupant protection and graceful degradation. It remains a design concept until the architecture is substantiated through controlled analysis, component characterization, structural testing, dynamic impact testing, propulsion testing, fault-injection/HIL testing, flight testing, fire testing, water-impact testing and the applicable certification process.

Next highest-value engineering closures:

- Freeze detailed mission profile and design operating envelope.
- Create controlled CAD geometry for the seven rotor stations and all seven landing legs.
- Complete rotor aerodynamic/interference and control-authority model.
- Close propulsion/generator/motor electrical and thermal architecture.
- Replace planning mass allocations with component-level data.
- Define landing-leg and capsule-damper force/stroke requirements.
- Perform formal FHA/FMEA/FMECA/FTA/PSSA/SSA work as applicable.
- Define emergency-recovery sizing and structural attachment loads.
- Define water-impact, flotation and egress requirements.
- Establish requirement/evidence database and verification matrix.

FINAL PRINCIPLE: ENGINEERING TRUTH > DOCUMENT COMPLETENESS. EVIDENCE > ASSERTION. CONFIGURATION CONTROL > SILENT CORRECTION. TRACEABILITY > CONVENIENCE.

Document-quality assessment: 9.7/10 as an engineering-document control baseline, conditional on preserving the evidence boundaries and closing the open technical items. This score is not a safety or feasibility certification.

Founder / Principal Author: Amirbahador Atai
Document: 00 | Revision: 00-T10-R7-9.7
Configuration: GEN3-SH-10T-FC | Baseline Revision: Rev A
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